Анализ модельного риска
дз7
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1. Delta $\dfrac{\partial V}{\partial S}$ for a geometric Asian call

In the modified Black–Scholes–Merton model with volatility estimated from a GARCH(1,1) model we assume that the GARCH forecast provides an effective (time–to–maturity) volatility $\hat{\sigma}_A$ for the geometric average over $[0, T]$. Under this assumption the price of a geometric Asian call can be written in Black–Scholes form

$$V(S) = Se^{-q_A T} N(d_1) - Ke^{-rT} N(d_2),$$

where

$$d_1 = \frac{\ln(S/K) + (r - q_A + \frac{1}{2}\hat{\sigma}_A^2)T}{\hat{\sigma}_A \sqrt{T}}, \qquad d_2 = d_1 - \hat{\sigma}_A \sqrt{T},$$

q_A is the effective yield coming from geometric averaging, and $N(\cdot)$ denotes the standard normal cdf. The GARCH(1,1) volatility $\hat{\sigma}_A$ is treated as given (it depends on past returns and time, not on the current S), so $\partial \hat{\sigma}_A / \partial S = 0$.

We differentiate V with respect to S. Using $\varphi(x) = N'(x)$ for the standard normal pdf, the chain rule gives

$$\frac{\partial V}{\partial S} = e^{-q_A T} N(d_1) + Se^{-q_A T} \varphi(d_1) \frac{\partial d_1}{\partial S} - Ke^{-rT} \varphi(d_2) \frac{\partial d_2}{\partial S}.$$

Because $\hat{\sigma}_A$ does not depend on S,

$$\frac{\partial d_1}{\partial S} = \frac{1}{S\, \hat{\sigma}_A \sqrt{T}}, \qquad \frac{\partial d_2}{\partial S} = \frac{\partial d_1}{\partial S} = \frac{1}{S\, \hat{\sigma}_A \sqrt{T}}.$$

Hence

$$\frac{\partial V}{\partial S} = e^{-q_A T} N(d_1) + e^{-q_A T} \frac{S\, \varphi(d_1)}{S\, \hat{\sigma}_A \sqrt{T}} - e^{-rT} \frac{K\, \varphi(d_2)}{S\, \hat{\sigma}_A \sqrt{T}}.$$

We now use the standard Black–Scholes identity

$$Se^{-q_A T} \varphi(d_1) = Ke^{-rT} \varphi(d_2)$$

(which follows from the relation $d_2 = d_1 - \hat{\sigma}_A \sqrt{T}$ and a direct computation of $\varphi(d_2)/\varphi(d_1)$). Therefore the two last terms cancel:

$$e^{-q_A T} \frac{S\, \varphi(d_1)}{S\, \hat{\sigma}_A \sqrt{T}} - e^{-rT} \frac{K\, \varphi(d_2)}{S\, \hat{\sigma}_A \sqrt{T}} = 0.$$

Thus the delta of the geometric Asian call in the modified BSM–GARCH framework is

$$\boxed{\frac{\partial V}{\partial S} = e^{-q_A T} N(d_1)}$$

which has the same form as for a vanilla call, but with the effective yield q_A and the GARCH–based volatility $\hat{\sigma}_A$ entering d_1, d_2.

2. Gamma $\dfrac{\partial^2 V}{\partial S^2}$ for a geometric Asian call

We can obtain gamma by differentiating the delta we just derived:

$$\Delta(S) \equiv \frac{\partial V}{\partial S} = e^{-q_A T} N(d_1).$$

Using again $\varphi = N'$ and the chain rule,

$$\Gamma(S) \equiv \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S} = e^{-q_A T}\varphi(d_1)\frac{\partial d_1}{\partial S}.$$

From above,

$$\frac{\partial d_1}{\partial S} = \frac{1}{S\,\hat{\sigma}_A\sqrt{T}},$$

so

$$\Gamma(S) = e^{-q_A T}\varphi(d_1)\frac{1}{S\,\hat{\sigma}_A\sqrt{T}}.$$

Therefore, the gamma of the geometric Asian call in the modified BSM–GARCH setup is

$$\boxed{\frac{\partial^2 V}{\partial S^2} = \frac{e^{-q_A T}\,\varphi(d_1)}{S\,\hat{\sigma}_A\sqrt{T}}}$$

again identical in form to the Black–Scholes gamma, with the GARCH-estimated effective volatility $\hat{\sigma}_A$ used in d_1.

3. Relation between $f(t, T_1)$ and $f(t, T_2)$ in the Hull–White model

In the one–factor Hull–White model the short rate satisfies

$$dr_t = (\theta - \kappa r_t)\,dt + \sigma\,dW_t,$$

with constant θ and σ. The zero–coupon bond price is

$$P(t,T) = \mathbb{E}_t\Big[\exp\Big(-\int_t^T r_u\,du\Big)\Big],$$

and, because $\int_t^T r_u\,du$ is Gaussian, this reduces to an affine form $P(t,T) = \exp(A(t,T)r_t + B(t,T))$ as discussed in Lecture 7.

Let $\tau = T - t$. Solving the OU process and computing the mean and variance of $\int_t^T r_u\,du$ one obtains

$$f(t,T) \;=\; -\frac{\partial}{\partial T}\ln P(t,T) \;=\; r_t e^{-\kappa\tau} + \frac{\theta}{\kappa}\big(1 - e^{-\kappa\tau}\big) - \frac{\sigma^2}{4\kappa}\big(1 - e^{-2\kappa\tau}\big),$$

i.e.

$$\boxed{f(t,T) = r_t e^{-\kappa(T-t)} + \frac{\theta}{\kappa}\big(1 - e^{-\kappa(T-t)}\big) - \frac{\sigma^2}{4\kappa}\big(1 - e^{-2\kappa(T-t)}\big)}$$

Now set $\tau_i = T_i - t$ for $i = 1, 2$ with $T_1 < T_2$. Then

$$f(t,T_i) = r_t e^{-\kappa\tau_i} + \frac{\theta}{\kappa}\big(1 - e^{-\kappa\tau_i}\big) - \frac{\sigma^2}{4\kappa}\big(1 - e^{-2\kappa\tau_i}\big), \qquad i = 1, 2.$$

Eliminating r_t from these two equations gives a direct relationship between the two instantaneous forward rates:

$$f(t,T_2) = e^{-\kappa(\tau_2-\tau_1)}\Big[f(t,T_1) - A(\tau_1)\Big] + A(\tau_2),$$

$$A(\tau) = \frac{\theta}{\kappa}\big(1 - e^{-\kappa\tau}\big) - \frac{\sigma^2}{4\kappa}\big(1 - e^{-2\kappa\tau}\big).$$

Equivalently,

$$\boxed{f(t,T_2) = e^{-\kappa(T_2-T_1)}f(t,T_1) + A(T_2 - t) - e^{-\kappa(T_2-T_1)}A(T_1 - t)}$$

which is the required affine relation between $f(t, T_1)$ and $f(t, T_2)$ in the Hull–White model with constant θ and σ.